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Addison Smith

CONTROLS / AI & ML ENGINEERING

Controls Application Engineer working on BMS and SCADA systems for hyperscale data centers and industrial plants, and the software around them. Ignition Gold certified, hands-on with Niagara, Ignition and Allen-Bradley. Currently pursuing a BS in Information Technology (AI/ML concentration) with a 4.0 GPA.

EXPERIENCE

Controls Application Engineer

Aug 2026 to Present

Mantis Innovation | Remote

DATA CENTER BMS / SCADA

- Design and application engineering for BMS and EPMS systems in hyperscale data centers
- Niagara N4 work across Supervisors and JACEs: device migrations, graphics, alarming and history configuration
- Typical flow diagrams and design documentation for data center mechanical and electrical systems
- Ignition SCADA development and support for industrial clients
- Python and scripting to automate repetitive engineering work

Controls Engineering Technician

Sep 2025 to Aug 2026

Daedalus Industrial | Easley, SC

CONTROLS ENGINEERING

- Program and commission PLC systems in Studio 5000 Logix Designer (CompactLogix/ControlLogix), including sequencer logic, PIDE loops, valve control, and pump control routines
- Design and deploy HMI interfaces in FactoryTalk View for operator monitoring and process control in regulated manufacturing environments
- Execute wiring, setup, and testing of control panels for automation and R&D projects, following UL508A and NEC safety standards
- Commission systems on site, including tank farm automation using Ethernet/IP and Modbus TCP protocols

MACHINE LEARNING & SOFTWARE DEVELOPMENT

- Built and deployed computer vision inspection systems for real-time defect detection in manufacturing (golf ball sorting, panel component verification)
- Developed an L5X generator that programmatically creates Rockwell Ladder Logic by ingesting existing L5X project files and building new programs from templates for direct import into Studio 5000
- Designed a multi-agent AI system (CrewAI/Ollama) for engineering automation, including vision analysis, code generation, and task orchestration across specialized teams
- Support data collection, labeling, and model training to improve detection accuracy across production lines

Electrical Technician

Jun 2024 to Sep 2025

AIRSYS USA | Anderson, SC

- Provided R&D support for cooling systems used in chip fabrication and data center applications, including two-phase direct-to-chip cooling testing
- Built and wired control panels operating from 110V to 480V for production and prototype units
- Read schematics, troubleshot electrical systems, and configured sensors during prototype testing
- Collaborated with engineers on prototype testing, system updates, and validation

Wiring Technician

Sep 2023 to Jun 2024

Vertiv | Anderson, SC

- Wired industrial switchgear systems (MV/LV, UPS, ATS) following detailed schematics and NEC safety codes
- Performed quality checks and documentation to ensure accuracy and reliability of panel assemblies
- Gained direct experience with data center critical power infrastructure and Vertiv product lines

PROJECTS

Chalk: Football Intelligence Platform

Building a coaching workspace with three modules (Film Room, Board, Playbook) using Next.js, React, TypeScript, Tailwind CSS, Supabase, Vercel

Smith_Agentic: Multi-Agent AI System

Designed a modular multi-agent architecture with 20 agents across four specialized crews (Default, PLC, React, Vision) built with CrewAI and Ollama

L5X Ladder Logic Generator

Python tool that ingests Rockwell L5X files, builds new PLC programs from templates, and exports for direct import into Studio 5000

RFID Race Timing System

UHF RFID timing for road races: Impinj R420 reader, LMR-400 antenna runs, and bib tag pairing for finish line results

TECHNICAL SKILLS

PROGRAMMING	Python, TypeScript, JavaScript, SQL, BQL, HTML/CSS, Ladder Logic, Structured Text, Function Block, Git, GitHub
AUTOMATION & CONTROLS	Ignition SCADA (Vision, Perspective), Niagara N4 (Vykon, EC-Net), Studio 5000 Logix Designer, FactoryTalk View, Allen-Bradley CompactLogix/ControlLogix, PLC Programming & Commissioning, UL508A Panel Build, Industrial Wiring (110V-480V), Visio
PROTOCOLS	BACnet IP and MS/TP, Modbus TCP/RTU, Ethernet/IP, OPC-UA, TCP/IP, Industrial Networking
DATA & AI	Machine Learning, Computer Vision, Defect Detection, Data Analysis, LLM APIs, Agentic Systems, Model Context Protocol (MCP)
FRAMEWORKS & PLATFORMS	React, Next.js, Tailwind CSS, FastAPI, Supabase, Vercel, CrewAI, LangChain, Ollama, Linux, Windows

EDUCATION

Bachelor of Science in Information Technology

Feb 2027 (Expected)

ECPI University | AI/ML & Data Analytics Track | 4.0 GPA

Associate of Science in Mechatronics, Robotics & Automation Engineering

Jan 2019

Piedmont Technical College | 3.8 GPA

CERTIFICATIONS

Ignition Gold Certified, Inductive Automation (Oct 2026)

Ignition Core Certified, Inductive Automation (May 2026)

Building RAG Agents with LLMs, NVIDIA Deep Learning Institute (Sep 2026)

Building Agentic AI Applications with LLMs, NVIDIA Deep Learning Institute (Apr 2026)

Fundamentals of Deep Learning, NVIDIA Deep Learning Institute (Feb 2026)

Foundation: Introduction to LangChain - Python, LangChain Academy (May 2026)

Model Context Protocol: Advanced Topics, Anthropic (May 2026)

Introduction to Model Context Protocol, Anthropic (Apr 2026)

Workbench Basics and Niagara Data Service, Tridium University (Aug 2026)

Expert Panel Contributor, Data Center Cooling, Mordor Intelligence (Sep 2025)